Turbulence forecasting · validation

Model: yalamanchi_vit | checkpoint epoch: 100

Training: rollout, horizon 5

Forecast: 10 recursive steps; no ground-truth resets

Start: file 451, frame 0, t=121.722496

End: t=122.022331 | elapsed simulation time: 0.299835

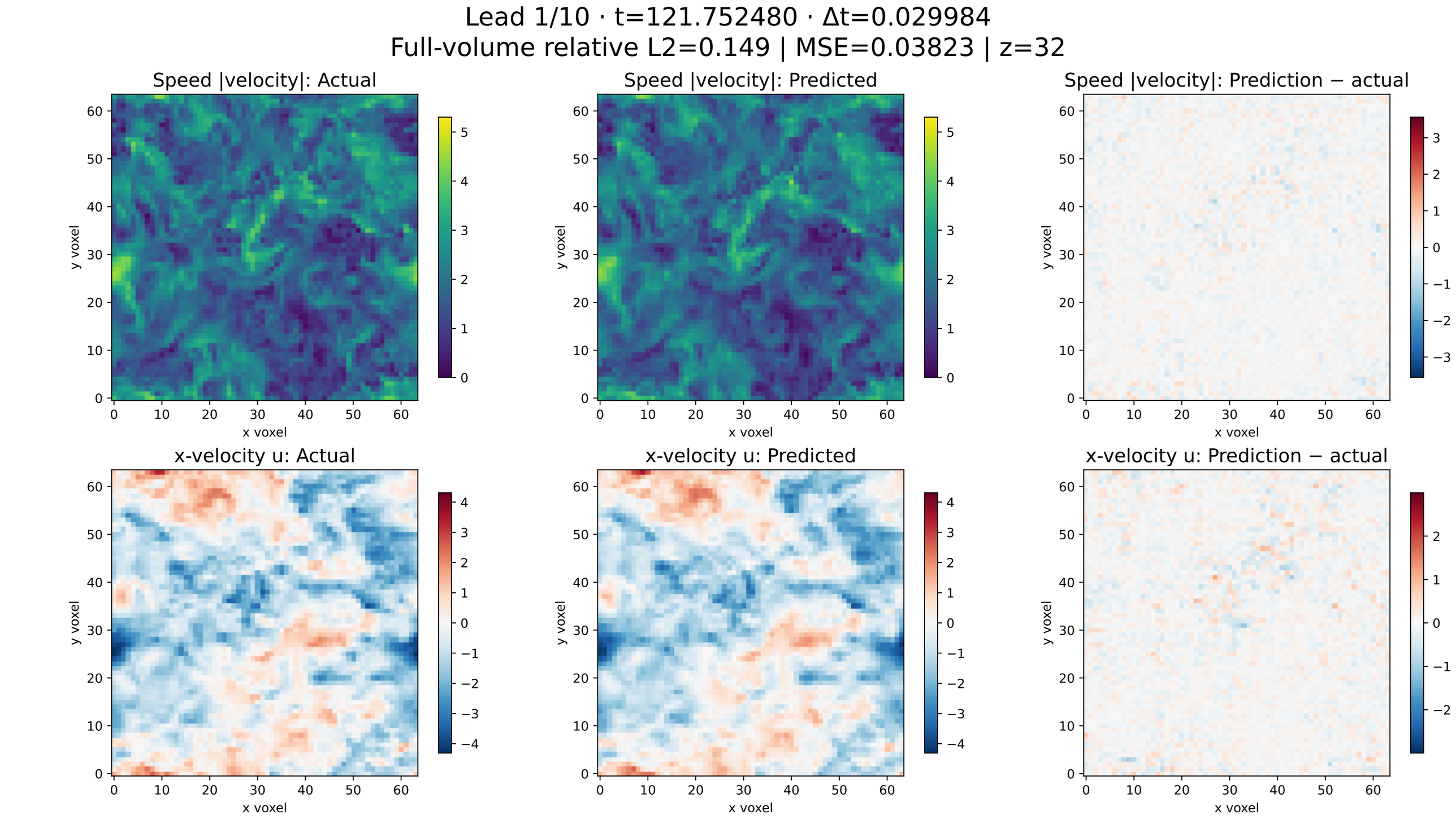
Filtered 64^3 velocity data | fixed z=32 slice

Fixed color scales across all leads; errors are prediction minus actual.

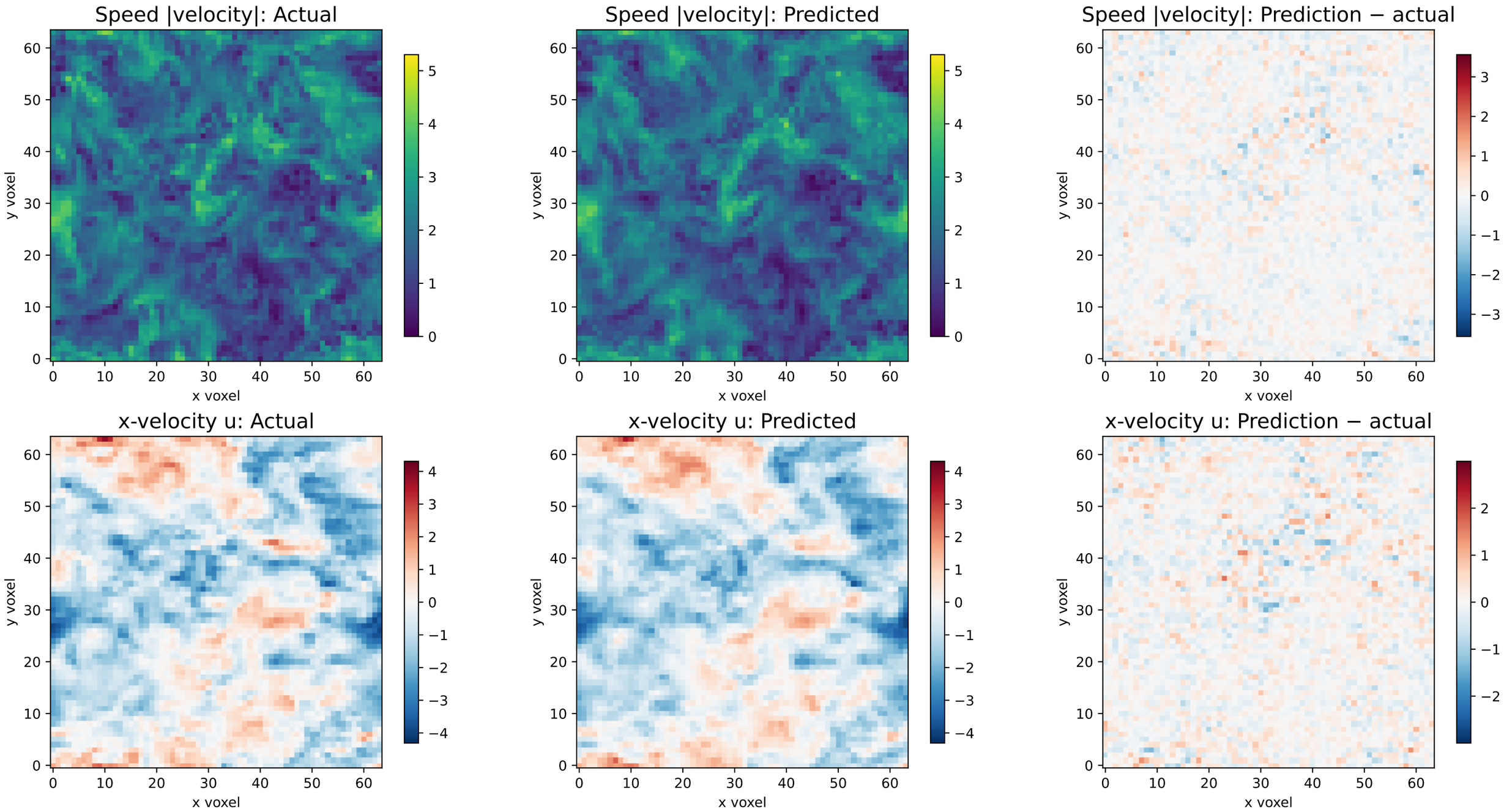
Field units are undocumented; metrics use original cached field values.

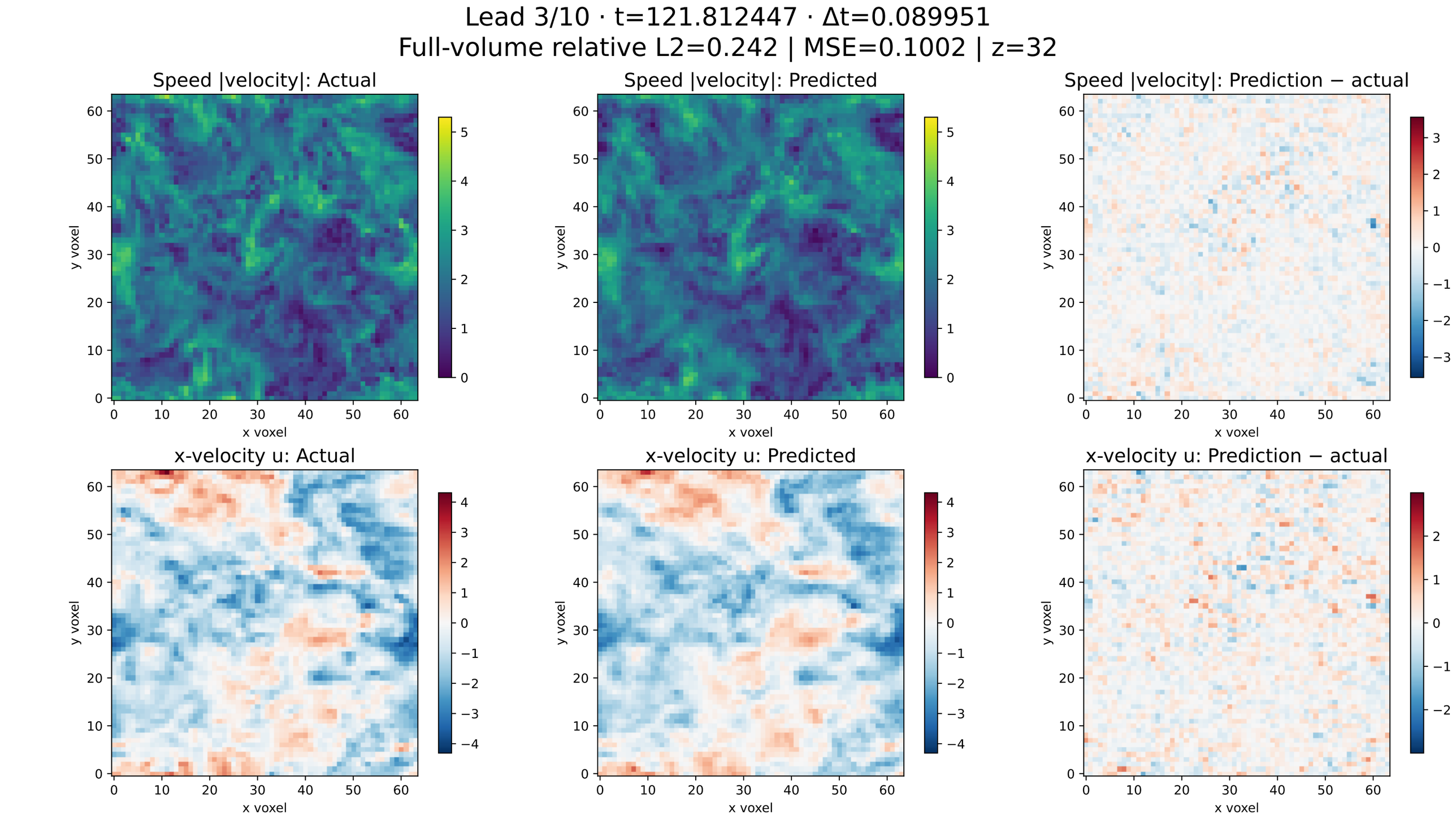
This trajectory is the first validation window, selected before evaluation.

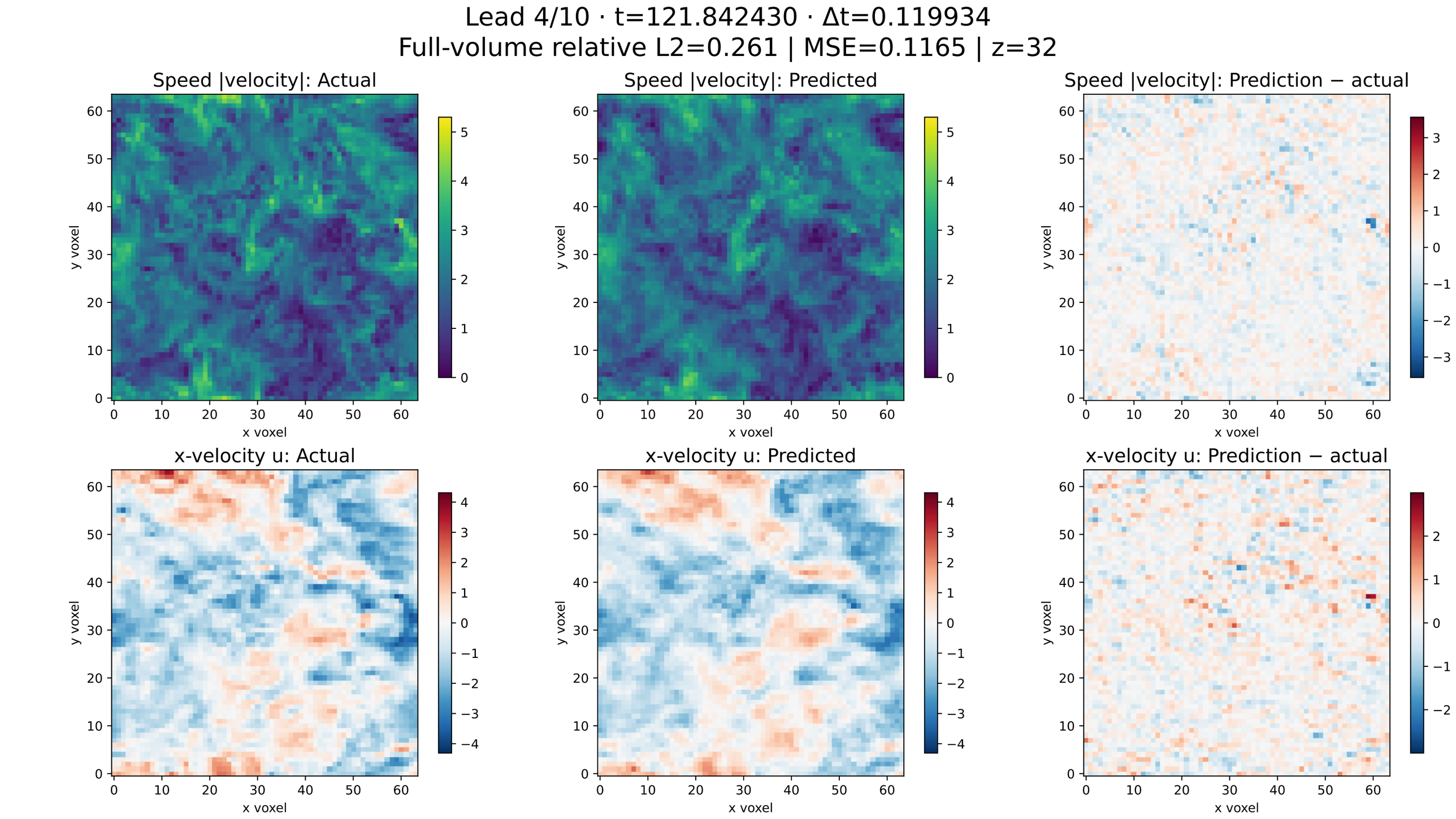
Validation was used for checkpoint selection; these are not independent test results.

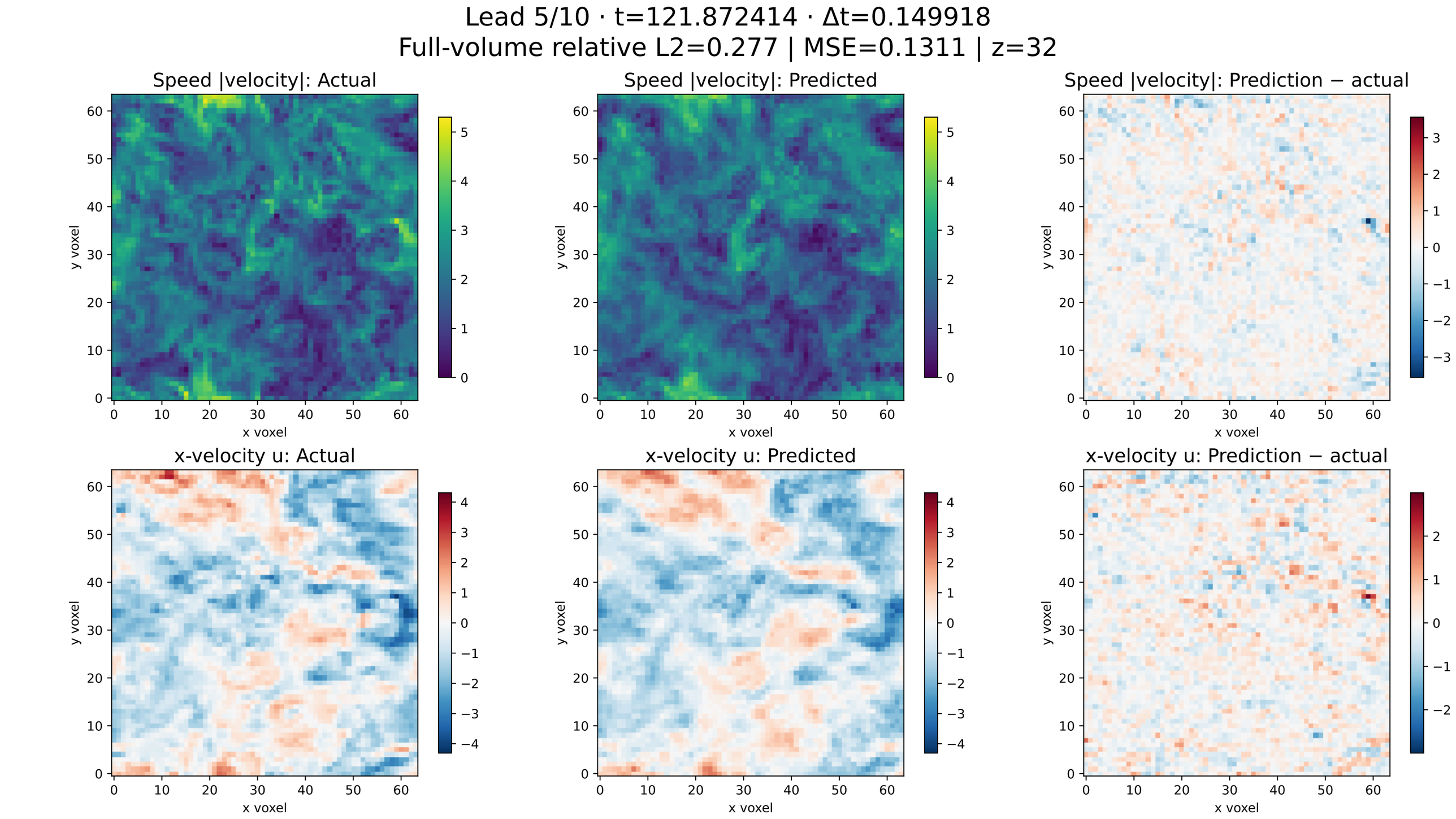

Lead 1/10 · t=121.752480 · Δt=0.029984
Full-volume relative L2=0.149 | MSE=0.03823 | z=32
Speed |velocity|: Actual
Speed |velocity|: Predicted
Speed |velocity|: Prediction − actual
x-velocity u: Actual
x-velocity u: Predicted
x-velocity u: Prediction − actual
x voxel
y voxel

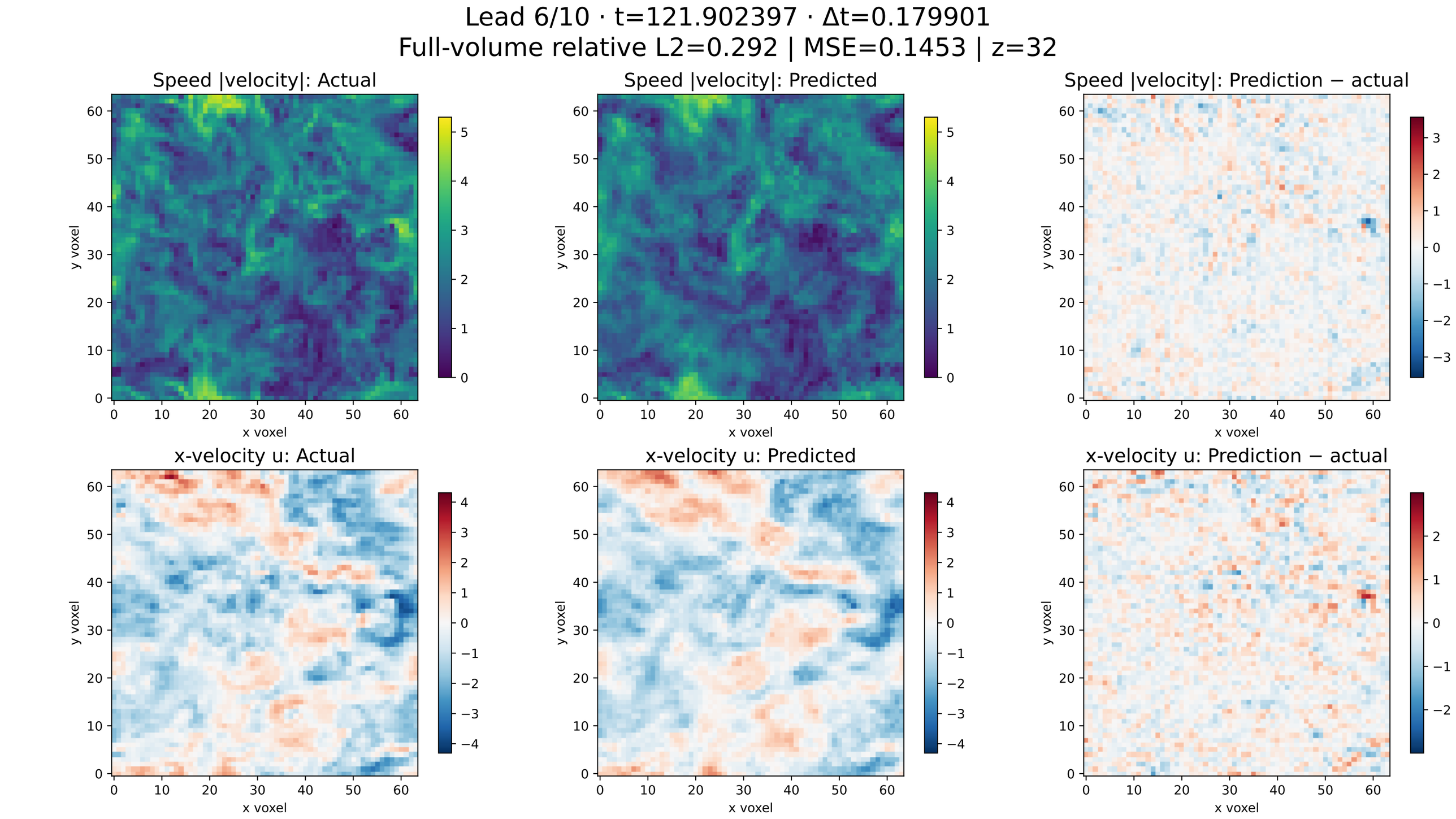
Lead 2/10 · t=121.782463 · Δt=0.059967
Full-volume relative L2=0.214 | MSE=0.07846 | z=32

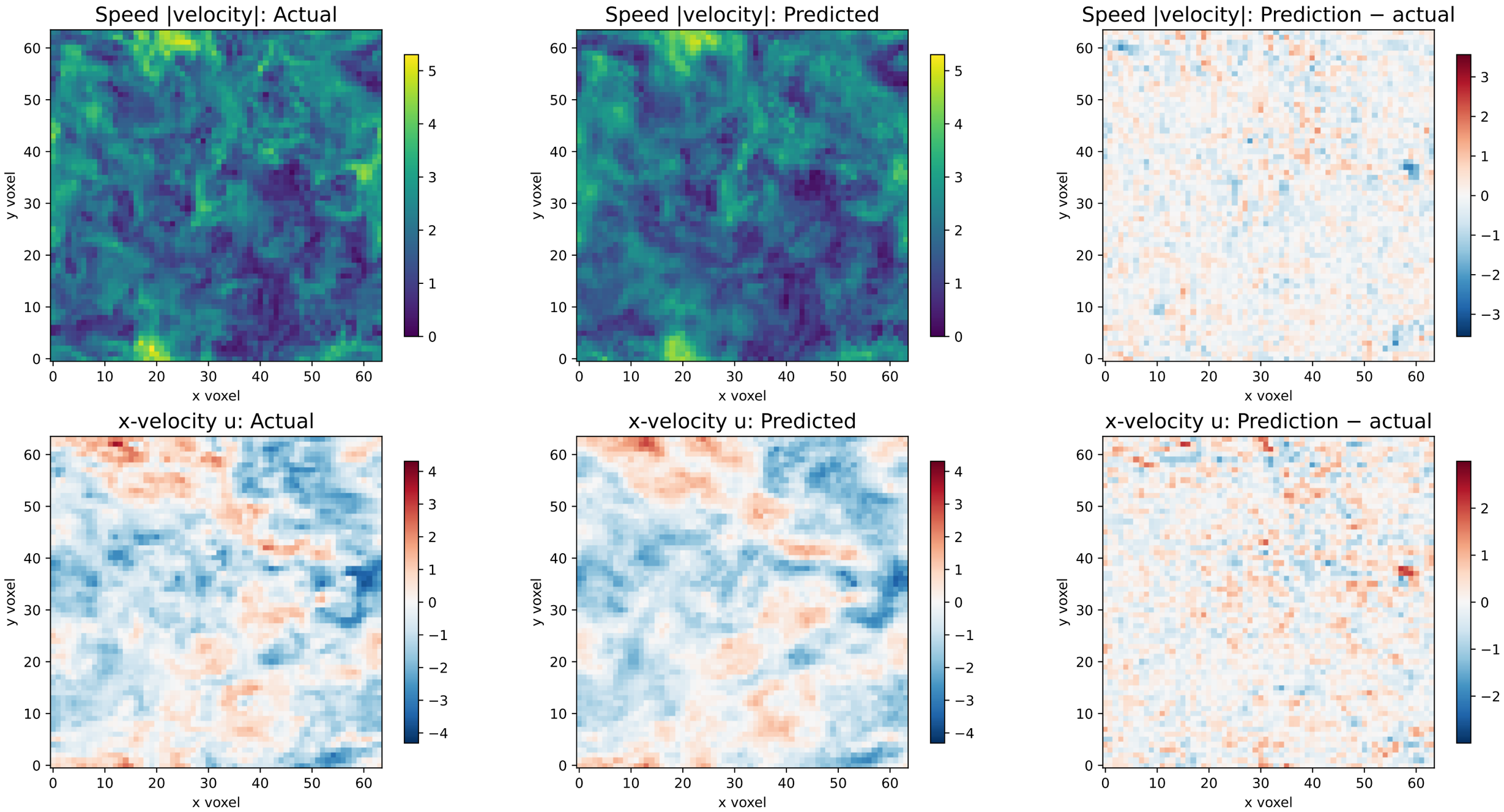



Lead 3/10 · t=121.812447 · Δt=0.089951
Full-volume relative L2=0.242 | MSE=0.1002 | z=32
Speed |velocity|: Actual
Speed |velocity|: Predicted
Speed |velocity|: Prediction − actual
x-velocity u: Actual
x-velocity u: Predicted
x-velocity u: Prediction − actual
x voxel
y voxel

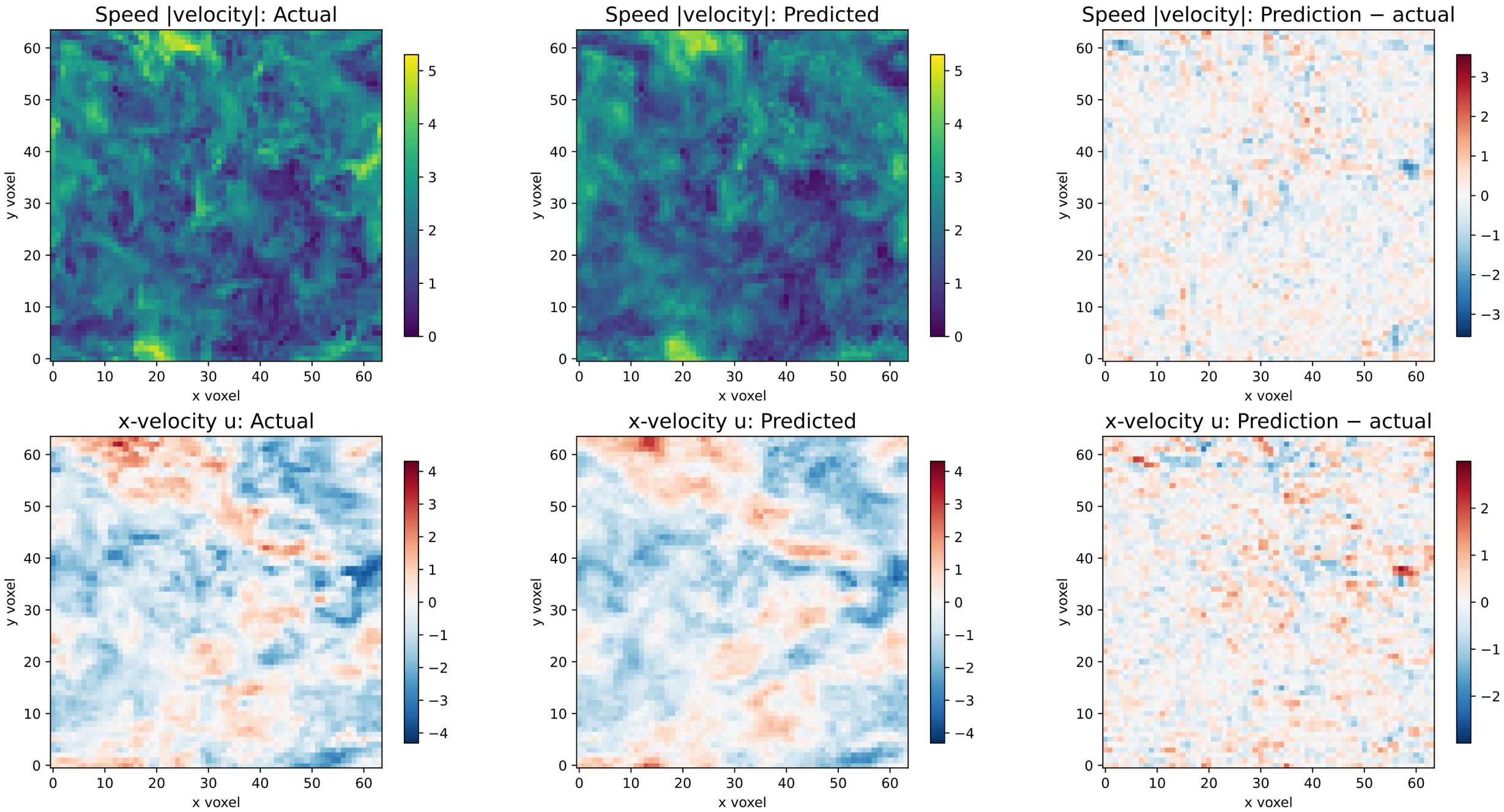

Lead 4/10 · t=121.842430 · Δt=0.119934
Full-volume relative L2=0.261 | MSE=0.1165 | z=32
Speed |velocity|: Actual
Speed |velocity|: Predicted
Speed |velocity|: Prediction − actual
x-velocity u: Actual
x-velocity u: Predicted
x-velocity u: Prediction − actual
x voxel
y voxel

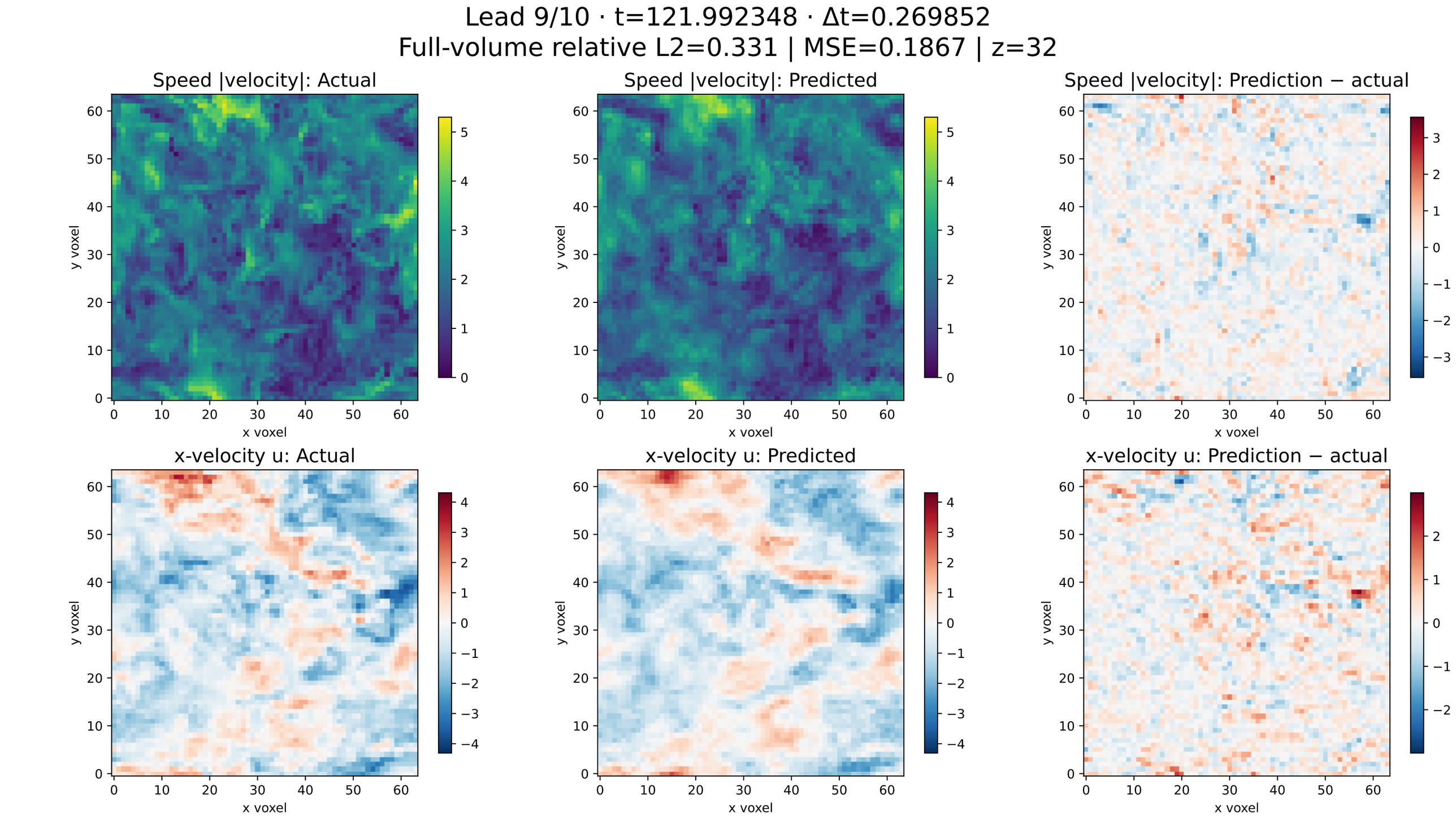

Lead 5/10 · t=121.872414 · Δt=0.149918
Full-volume relative L2=0.277 | MSE=0.1311 | z=32
Speed |velocity|: Actual
Speed |velocity|: Predicted
Speed |velocity|: Prediction − actual
x-velocity u: Actual
x-velocity u: Predicted
x-velocity u: Prediction − actual
x voxel
y voxel

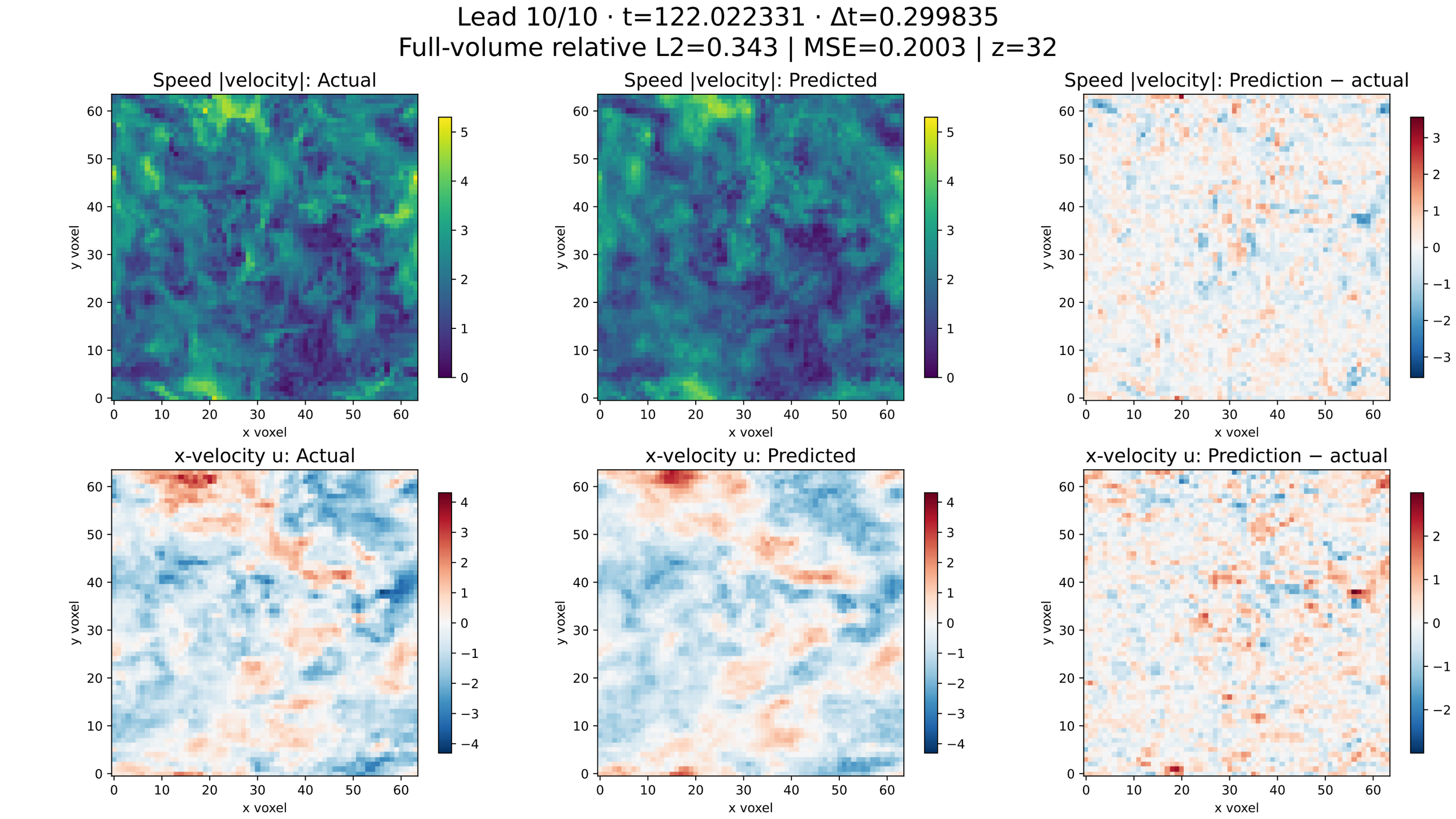

Lead 6/10 · t=121.902397 · Δt=0.179901
Full-volume relative L2=0.292 | MSE=0.1453 | z=32
Speed |velocity|: Actual
Speed |velocity|: Predicted
Speed |velocity|: Prediction − actual
x-velocity u: Actual
x-velocity u: Predicted
x-velocity u: Prediction − actual
y voxel
x voxel

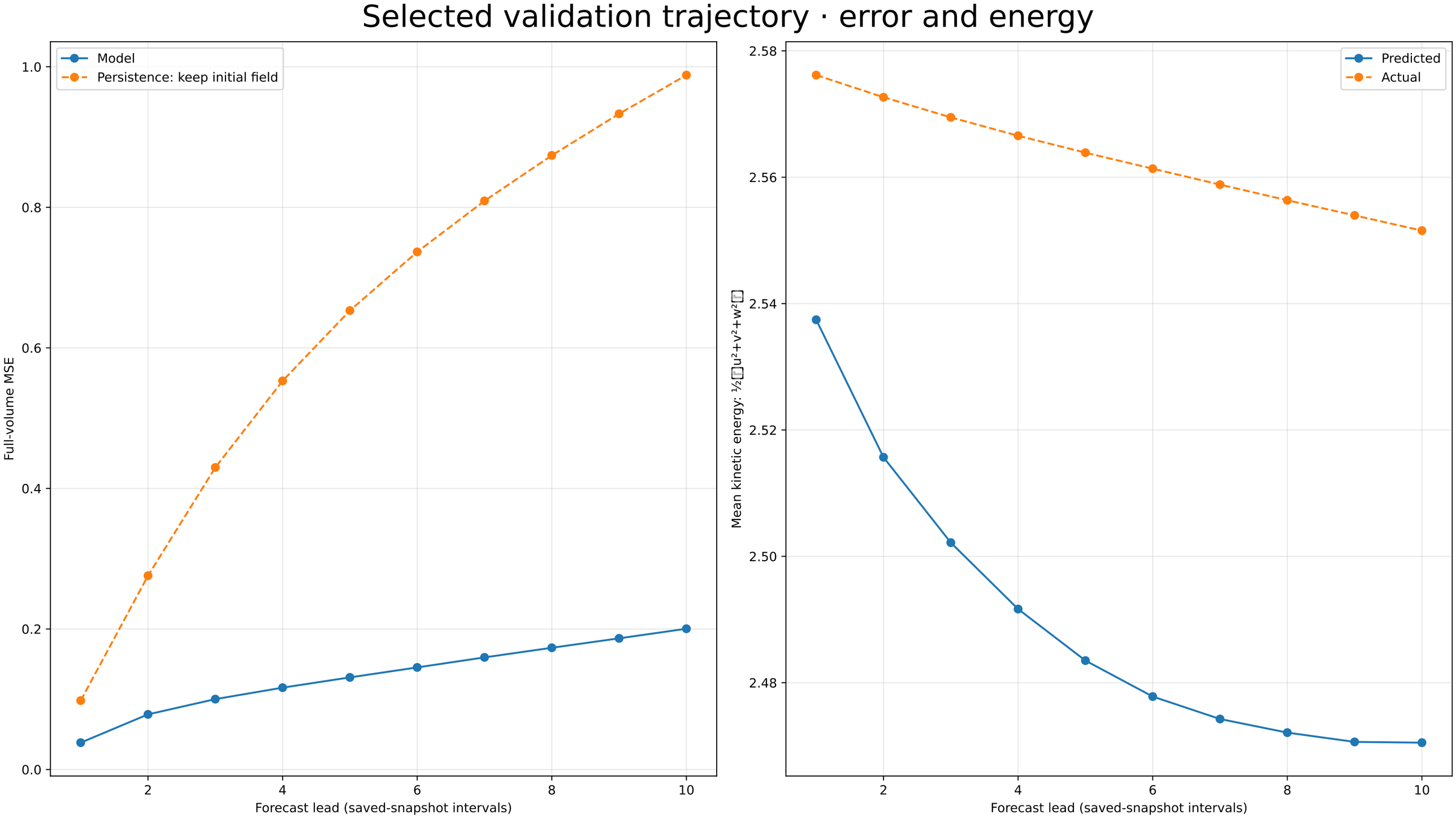

Lead 7/10 · t=121.932381 · Δt=0.209885
Full-volume relative L2=0.306 | MSE=0.1596 | z=32
Speed |velocity|: Actual
Speed |velocity|: Predicted
Speed |velocity|: Prediction − actual
x-velocity u: Actual
x-velocity u: Predicted
x-velocity u: Prediction − actual
y voxel
x voxel

Lead 8/10 · t=121.962364 · Δt=0.239868
Full-volume relative L2=0.319 | MSE=0.1732 | z=32




Lead 9/10 · t=121.992348 · Δt=0.269852
Full-volume relative L2=0.331 | MSE=0.1867 | z=32
Speed |velocity|: Actual
Speed |velocity|: Predicted
Speed |velocity|: Prediction − actual
x-velocity u: Actual
x-velocity u: Predicted
x-velocity u: Prediction − actual
x voxel
y voxel


Lead 10/10 · t=122.022331 · Δt=0.299835
Full-volume relative L2=0.343 | MSE=0.2003 | z=32
Speed |velocity|: Actual
Speed |velocity|: Predicted
Speed |velocity|: Prediction − actual
x-velocity u: Actual
x-velocity u: Predicted
x-velocity u: Prediction − actual
x voxel
y voxel


Selected validation trajectory · error and energy
Model
Persistence: keep initial field
Full-volume MSE
Forecast lead (saved-snapshot intervals)
Predicted
Actual
Mean kinetic energy: ½∫(u²+v²+w²)
Forecast lead (saved-snapshot intervals)